

Math 220A HW 2

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1 D

Problem 1

Prove that all finite p -groups are solvable.

Solution: Given any finite p -group G (where p is prime), the order $|G| = p^k$ for some $k \in \mathbb{N} \setminus \{0\}$. Hence, we'll perform induction on the exponent k .

For base case $k = 1$, if $|G| = p$, then since p is prime then $G \cong \mathbb{Z}/p\mathbb{Z}$ as group (must be cyclic, hence abelian). Then, since abelian groups are vacuously solvable, G is solvable.

Now, suppose for given $n \in \mathbb{N}$, any p -group G with $|G| = p^k$ (where $k < n$) is solvable. Then, for the case of any p -group H with $|H| = p^n$, we'll consider the following statements:

1. **H has nontrivial Center:**

If consider the conjugation action of H on itself, then all the singletons among the conjugation classes are precisely the elements in the center (since $\{h\} \subseteq H$ is a conjugation class $\iff ghg^{-1} = h$ for all $g \in H \iff gh = hg$ for all $g \in H$, which is characterized as $h \in Z(H)$, the center of H). Then based on the class equation, we get the following:

$$|H| = |Z(H)| + \sum_{j \in J} |\text{Conj}(h_j)| \quad (1.1)$$

Where the list of J runs through distinct $h_j \in H$, each represent a distinct conjugation class that is not singleton (and $\text{Conj}(h_j)$ denotes the conjugation class of h_j). Then, since by Orbit Stabilizer Theorem, one has $|H| = p^n$ being divisible by $|\text{Conj}(h_j)|$ (which is not 1), then one must have $|\text{Conj}(h_j)| = p^{i_j}$ for some integer $i_j > 0$, so $|\text{Conj}(h_j)|$ is divisible by p . Hence, every element in the sum $\sum_{j \in J} |\text{Conj}(h_j)|$ are divisible by p , $\sum_{j \in J} |\text{Conj}(h_j)| = p \cdot k$ for some integer $k \in \mathbb{Z}$. Together with $|H| = p^n$, one has the following:

$$|Z(H)| = |H| - \sum_{j \in J} |\text{Conj}(h_j)| = p^n - p \cdot k = p \cdot (p^{n-1} - k) \quad (1.2)$$

This shows that $|Z(H)|$ is divisible by p , hence $|Z(H)| \neq 1$, showing the center is nontrivial.

2. **Preimage of Group Homomorphism preserves Normality:**

Given a group homomorphism $f : K \twoheadrightarrow L$, let $N \leq M \leq L$ be subgroups such that $N \trianglelefteq M$. If consider their preimage $N' = f^{-1}(N)$ and $M' = f^{-1}(M)$ (where $N' \leq M'$), one has $f(N') \leq N$ and $f(M') \leq M$. Now, given any $r \in N'$ and $q \in M'$, if consider qrq^{-1} , we have the following:

$$f(qrq^{-1}) = f(q)f(r)f(q)^{-1} \in N \quad (1.3)$$

Since $f(q) \in M$ and $f(r) \in N$, based on normality $N \trianglelefteq M$ one has $f(q)f(r)f(q)^{-1} \in N$. Therefore, $qrq^{-1} \in f^{-1}(N) = N'$, showing that $N' \trianglelefteq M'$.

As a consequence of **1**, with $|Z(H)|$ divides $|H| = p^n$, one must have $|Z(H)| = p^k$ for some integer $0 < k \leq n$; together with $Z(H)$ being abelian, utilizing Cauchy's Theorem, there exists element $l \in Z(H)$ such that $|l| = p$, so $\langle l \rangle \subseteq H$ is a central (hence normal) subgroup of H with order p . Then, $H/\langle l \rangle$ is a quotient group with $|H/\langle l \rangle| = \frac{p^n}{p} = p^{n-1}$ also being a p -group (or a trivial group, if $n = 1$). Hence, with $(n-1) < n$, by induction $H/\langle l \rangle$ is solvable. Hence, there exists chain/tower of subgroups $\{e\} = K_0 < K_1 < \dots < K_m = H/\langle l \rangle$, such that for each index i , $K_i \trianglelefteq K_{i+1}$, and K_{i+1}/K_i is abelian.

Now, using the condition given by **2**, pullback the tower using the projection $\pi : H \twoheadrightarrow H/\langle l \rangle$, let $K'_i := \pi^{-1}(K_i)$, it admits the following tower:

$$\{e\} < \langle l \rangle = K'_0 < K'_1 < \dots < K'_m = H \quad (1.4)$$

Since π is surjective, then $K'_m = \pi^{-1}(K_m) = \pi^{-1}(H/\langle l \rangle)$; also, since $\pi^{-1}(K_0) = \pi^{-1}(\{1\}) = \ker(\pi)$, with π being projection, then $K'_0 = \pi^{-1}(\{1\}) = \ker(\pi) = \langle l \rangle$. And, the reason this is a tower, is because $K_i \trianglelefteq K_{i+1}$ implies $K'_i \trianglelefteq K'_{i+1}$ by consequence of **2**.

Finally, to prove that such tower is abelian, for each index i , the projection restricts to $\pi : K'_i \twoheadrightarrow K_i$ (since π is surjective, so $\pi(K'_i) = \pi(\pi^{-1}(K_i)) = K_i$). Now, given the restriction $\pi : K'_{i+1} \twoheadrightarrow K_{i+1}$, since $K_i \trianglelefteq K_{i+1}$, and K_{i+1}/K_i is abelian by assumption, then composing with the projection $\rho : K_{i+1} \twoheadrightarrow K_{i+1}/K_i$, one has $\rho \circ \pi : K'_{i+1} \twoheadrightarrow K_{i+1}/K_i$ being an surjective group homomorphism with abelian image, showing that $K'_{i+1}/\ker(\rho \circ \pi) \cong K_{i+1}/K_i$ being abelian also.

Now, notice that $\ker(\rho \circ \pi) = K'_i$: Given any $g \in K'_{i+1}$, one has $g \in \ker(\rho \circ \pi) \iff \rho(\pi(g)) = 1 \in K_{i+1}/K_i \iff \pi(g) \in K_i \iff g \in \pi^{-1}(K_i) = K'_i$. Hence, we in fact has K'_{i+1}/K'_i also being abelian, concluding that the following tower is abelian:

$$\{e\} < K'_0 < K'_1 < \dots < K'_m = H \quad (1.5)$$

(Note: the reason why $\{e\} < K'_0$ is part of the abelian tower, is simply because $K'_0 = \langle l \rangle$ is central, hence automatically abelian).

Hence, H (with $|H| = p^n$) is solvable. This finishes the induction.

2 D (Broken Down to Problem 2, 3)

Problem 2

Lang Algebra Chapter 1 #49:

A group as an automorphism group. Let G be a group and let $\mathbf{Set}(G)$ be the category of G -sets (i.e. sets with a G -operation / G -action). Let $F : \mathbf{Set}(G) \rightarrow \mathbf{Set}$ be the forgetful functor, which to each G -set assigns the set itself. Show that $\text{Aut}(F)$ is naturally isomorphic to G .

Solution: Given that $\text{Aut}(F)$ denotes the set of all natural isomorphisms from functor F to itself, to show it can be naturally isomorphic to G , we'll first claim some statements. As a side note, here $G \in \mathbf{Set}(G)$ is identified with its natural left action on itself (i.e. left multiplication), if any other action of G on itself shows up, it'll be specified.

1. **Any G -set X satisfies $\text{Hom}_{\mathbf{Set}(G)}(G, X) \cong X$ as sets:**

First, for any $f \in \text{Hom}_{\mathbf{Set}(G)}(G, X)$ (which preserves left G -action), it's uniquely identified by $f(e) \in X$. Since given any $g \in G$, one has $f(g) = f(g \cdot e) = g \cdot f(e)$, so knowing $f(e) \in X$ automatically determines the whole map f .

Conversely, for any $x \in X$, one can construct such map $f_x \in \text{Hom}_{\mathbf{Set}(G)}(G, X)$ such that $f_x(e) = x$, since define $f_x : G \rightarrow X$ by $f_x(g) := g \cdot x$ for all $g \in G$, satisfies $f_x(e) = e \cdot x = x$, and $g \cdot f_x(e) = g \cdot x = f_x(g) = f_x(g \cdot e)$, then f_x is a G -equivariant map with $f_x(e) = x$, showing that $f_x \in \text{Hom}_{\mathbf{Set}(G)}(G, X)$.

Hence, consider the map $\text{Hom}_{\mathbf{Set}(G)}(G, X) \rightarrow X$ by $f \mapsto f(e)$, this defines a natural bijection (since each $x \in X$ is endowed with a unique map $f_x \in \text{Hom}_{\mathbf{Set}(G)}(G, X)$ such that $f_x(e) = x$).

2. **Natural Isomorphism on F satisfies $\text{Aut}(F) \cong G$ as sets:**

Let $\eta : F \rightarrow F$ be a natural isomorphism (which, its inverse natural transformation $\eta^{-1} : F \rightarrow F$ satisfies the property that for all G -set $X \in \mathbf{Set}(G)$, $\eta_X : F(X) \rightarrow F(X)$ is invertible, and $(\eta_X)^{-1} = \eta_X^{-1}$).

Now, since F is just forgetful functor, as set $F(G) = G$, and for any $X \in \mathbf{Set}(G)$, all $f \in \text{Hom}_{\mathbf{Set}(G)}(G, X)$ has $Ff = f \in \text{Hom}_{\mathbf{Set}}(G, X)$ (which simply forgets the G -equivariance structure, as set maps they're the same). Notice that given any $X \in \mathbf{Set}(G)$, one has $\eta_X : F(X) \rightarrow F(X)$ (or $\eta_X : X \rightarrow X$ as set map) being completely determined by $\eta_G(e)$ (where $\eta_G : F(G) \rightarrow F(G)$, or $\eta_G : G \rightarrow G$ as set maps).

First, we'll show that $\eta_G(e) \in G$ completely determines the whole natural isomorphism: Given any $X \in \mathbf{Set}(G)$ and any $x \in X$, recall that from 1 there exists a unique G -equivariant map $f_x \in \text{Hom}_{\mathbf{Set}(G)}(G, X)$ such that $f_x(e) = x$. Which, η must satisfy the following:

$$\begin{array}{ccc} G & \xrightarrow{f_x} & X \\ \eta_G \downarrow & & \downarrow \eta_X \\ G & \xrightarrow{f_x} & X \end{array}$$

Hence, we get the following relation (given that $\eta_G(e) \in G$):

$$\eta_X(x) = \eta_X(f_x(e)) = f_x(\eta_G(e)) = f_x(\eta_G(e) \cdot e) = \eta_G(e) \cdot f_x(e) = \eta_G(e) \cdot x \quad (2.1)$$

Hence, given a natural isomorphism $\eta : F \rightarrow F$, it's completely determined by its assignment of e , or what $e \mapsto \eta_G(e)$ is.

Next, given any $g \in G$, there exists such natural isomorphism $\rho : F \xrightarrow{\sim} F$ such that $\rho_G(e) = g$: Given any $X \in \mathbf{Set}(G)$, define the corresponding morphism $\rho_X : X \xrightarrow{\sim} X$ as $\rho_X(x) = g \cdot x$. Since the G -action is always invertible, it's clear that ρ_X must be set isomorphism. So, the collection $\rho = \{\rho_X : X \xrightarrow{\sim} X \mid X \in \mathbf{Set}(G)\}$ has an „inverse“ collection (since each morphism has an inverse). So, to prove it's a natural isomorphism on F , it suffices to prove that ρ is a natural transformation (since a natural transformation is an isomorphism iff every morphism is an isomorphism).

Which, for any $X, Y \in \mathbf{Set}(G)$, given any $f \in \text{Hom}_{\mathbf{Set}(G)}(X, Y)$, any $x \in X$ and $h \in G$ satisfies $f(h \cdot x) = h \cdot f(x) \in Y$. Then, the collection of isomorphism ρ does the following job:

$$\rho_Y(f(x)) = g \cdot f(x) = f(g \cdot x) = f(\rho_X(x)) \quad (2.2)$$

Hence, this demonstrates that $\rho_Y \circ f = f \circ \rho_X$. So, the following diagram commutes for any such choice of $X, Y \in \mathbf{Set}(G)$, and G -equivariant map $f \in \text{Hom}_{\mathbf{Set}(G)}(X, Y)$:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \rho_X \downarrow & & \downarrow \rho_Y \\ X & \xrightarrow{f} & Y \end{array}$$

This shows that ρ is in fact a natural transformation on F the forgetful functor, and with every corresponding morphism being an isomorphism, it's a natural isomorphism.

Therefore, the assignment $\varphi : \text{Aut}(F) \rightarrow G$ by $\eta \mapsto \eta_G(e)$ defines a bijection (since each $g \in G$ has a unique $\eta \in \text{Aut}(F)$ such that $\eta_G(e) = g$).

With the map $\varphi : \text{Aut}(F) \xrightarrow{\sim} G$ being a bijection, to prove that $\text{Aut}(F) \cong G$ as groups, it suffices to show that φ is indeed a group homomorphism. However, given any $\rho, \eta \in \text{Aut}(F)$, we have $(\rho \circ \eta)_G = \rho_G \circ \eta_G$. Let $g = \rho_G(e)$ and $h = \eta_G(e)$, the relation proven in **2** shows that for all $k \in G$, one has $\rho_G(k) = \rho_G(e) \cdot k = gk$ (recall that here G is using left action on itself). Hence, the following relation is true:

$$(\rho \circ \eta)_G(e) = \rho_G \circ \eta_G(e) = \rho_{G(h)} = gh \quad (2.3)$$

Hence, since $\varphi(\rho) = \rho_G(e) = g$ and $\varphi(\eta) = \eta_G(e) = h$, we have $\varphi(\rho \circ \eta) = (\rho \circ \eta)_G(e) = gh = \varphi(\rho)\varphi(\eta)$. So, φ is indeed a group homomorphism, hence it's a group isomorphism between $\text{Aut}(F)$ and G . So, $\text{Aut}(F)$ and G are indeed isomorphic.

Problem 3

Lang Algebra Chapter 1 #53:

Let H, G, G' be groups, and let $f : H \rightarrow G$, $g : H \rightarrow G'$ be two homomorphisms. Define the notion of coproduct of these two homomorphisms over H , and show that it exists.

Solution: Here, we'll first define the *Coslice Category* over H (denoted as $\mathbf{Grp}^H$). Here, define $\text{Ob}(\mathbf{Grp}^H)$ to be the collection of all pair $(K, h_K : H \rightarrow K)$, where K is a group and h_K is a group homomorphism. Also, given any groups K, L with $h_K : H \rightarrow K$ and $h_L : H \rightarrow L$, define the set of morphisms $\text{Hom}_{\mathbf{Grp}^H}(h_K, h_L) := \{f : K \rightarrow L \mid h_L = f \circ h_K\}$, which are all the group homomorphism $f : K \rightarrow L$ such that the following diagram commutes:

$$\begin{array}{ccc} & H & \\ h_K \swarrow & & \searrow h_L \\ K & \xrightarrow{f} & L \end{array}$$

To show that such definitions form a category, it suffices to check that composition of morphisms is well-defined, and the endomorphism set contains identity (since associativity naturally follows from the associativity of set function composition).

First, given groups K, L, P and h_K, h_L, h_P three morphisms in $\mathbf{Grp}^H$, if $f \in \text{Hom}_{\mathbf{Grp}^H}(h_K, h_L)$ and $g \in \text{Hom}_{\mathbf{Grp}^H}(h_L, h_P)$, the following diagram commutes:

$$\begin{array}{ccccc} & & H & & \\ & h_K \swarrow & \downarrow h_L & \searrow h_P & \\ K & \xrightarrow{f} & L & \xrightarrow{g} & P \end{array}$$

This shows that $g \circ f$ is a morphism from h_K to h_P , or $g \circ f \in \text{Hom}_{\mathbf{Grp}^H}(h_K, h_P)$, hence composition is well-defined.

Finally, extending from the above notations, the identity $\text{id}_L : L \rightarrow L$ satisfies the following:

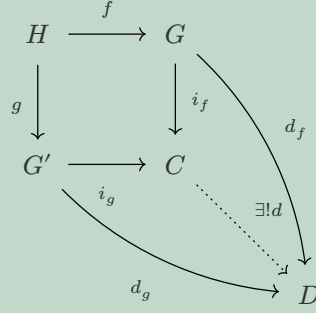
$$\begin{array}{ccccccc} & & H & & H & & \\ & h_K \swarrow & \downarrow h_L & \searrow h_L & h_L \swarrow & \downarrow h_L & \searrow h_P \\ K & \xrightarrow{f} & L & \xrightarrow{\text{id}_L} & L & \xrightarrow{\text{id}_L} & L \xrightarrow{g} P \end{array}$$

Hence, showing that $\text{id}_L \in \text{End}_{\mathbf{Grp}^H}(h_L)$, together with $\text{id}_L \circ f = f$ and $g \circ \text{id}_L = g$ (given $f \in \text{Hom}_{\mathbf{Grp}^H}(h_K, h_L)$ and $g \in \text{Hom}_{\mathbf{Grp}^H}(h_L, h_P)$). Hence, identity is well-defined, showing it's indeed a category.

To define a notion of coproduct of $f : H \rightarrow G$ and $g : H \rightarrow G'$ given in the problem, it'll be a group C with group homomorphism $h_C : H \rightarrow C$, together with two morphisms $i_f \in \text{Hom}_{\mathbf{Grp}^H}(f, h_C)$ and $i_g \in \text{Hom}_{\mathbf{Grp}^H}(g, h_C)$, such that $i_f \circ f = i_g \circ g$, and it is initial with this property.

Which means, given any other $h_D : H \rightarrow D$ in the category that's paired up with morphisms $d_f \in \text{Hom}_{\mathbf{Grp}^H}(f, h_D)$ and $d_g \in \text{Hom}_{\mathbf{Grp}^H}(g, h_D)$, there exists a unique morphism $d : C \rightarrow D$, such that $d_f = d \circ i_f$ and $d_g = d \circ i_g$.

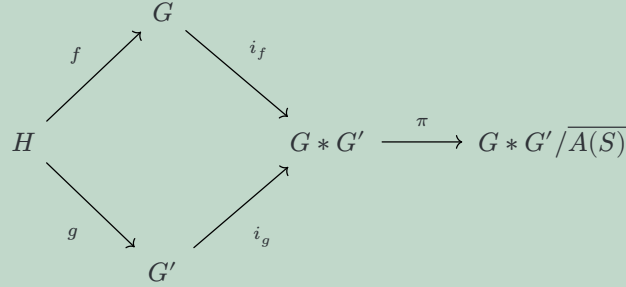
Which, by description one must have $h_C = i_f \circ f = i_g \circ g$, and $h_D = d_f \circ f = d_g \circ g$, so one can implicitly assume their existence in the diagram. Which, it becomes the following:



Hence, the object C together with morphism $i_f : G \rightarrow C$ and $i_g : G' \rightarrow C$ (which are groups together with two group homomorphisms into it) can equivalently be interpreted as fibre coproduct of $f : H \rightarrow G$ and $g : H \rightarrow G'$.

The last step is to show the existence. Since in **Grp** one knows the existence of coproduct, a free product of the two groups together with canonical injection. So, given $f : H \rightarrow G$ and $g : H \rightarrow G'$, consider the free product $G * G'$ and the two canonical injections $i_f : G \rightarrow G * G'$ and $i_g : G' \rightarrow G * G'$, we have two group homomorphisms $i_f \circ f, i_g \circ g : H \rightarrow G * G'$.

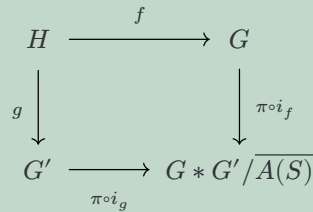
Then, one can take a coequalizer of the two group homomorphisms: Consider the set $S = \{i_f \circ f(x) \cdot (i_g \circ g(x))^{-1} \mid x \in H\} \subset G * G'$ (i.e. elements in image of $i_f \circ f$, conjugates with the corresponding image in $i_g \circ g$ based on what they get mapped from in H). Take the subgroup generated by it, say $A(S) \subseteq G * G'$, and take its normal closure $\overline{A(S)} \subseteq G * G'$ (or, the smallest normal subgroup containing $A(S)$). Consider the projection $\pi : G * G' \rightarrow G * G' / \overline{A(S)}$. So, now one has the following diagram (which the left part is not necessarily commutative):



Now, given the two group homomorphisms $\pi \circ i_f : G \rightarrow G * G' / \overline{A(S)}$ and $\pi \circ i_g : G' \rightarrow G * G' / \overline{A(S)}$, notice that given any $x \in H$, one has the following:

$$(\pi \circ i_f) \circ f(x) = \overline{i_f \circ f(x)} = \overline{i_g \circ g(x)} = (\pi \circ i_g) \circ g(x) \in G * G' / \overline{A(S)} \quad (2.4)$$

By construction, one has $(i_f \circ f(x)) \cdot (i_g \circ g(x))^{-1} \in \overline{A(S)}$, hence $\overline{i_f \circ f(x)} = \overline{i_g \circ g(x)}$ when taking the quotient. Hence, the relation is true, and reduces to the following diagram:



Now, given any other group D with two group homomorphisms $d_f : G \rightarrow D$ and $d_g : G' \rightarrow D$ satisfying $d_g \circ g = d_f \circ f$, by the definition, since $G * G'$ (together with $i_f : G \rightarrow G * G'$ and $i_g : G' \rightarrow G * G'$) is a coproduct of G, G' , there exists a unique group homomorphism $d : G * G' \rightarrow D$, such that $d_f = d \circ i_f$ and $d_g = d \circ i_g$. Or, the following diagram is true:

$$\begin{array}{ccccc}
 G & \xrightarrow{i_f} & G * G' & \xleftarrow{i_g} & G' \\
 & \searrow d_f & \downarrow \exists! d & \swarrow d_g & \\
 & & D & &
 \end{array}$$

Then, this indicates that $(d \circ i_f) \circ f = d_f \circ f = d_g \circ g = (d \circ i_g) \circ g$, in other words for all $x \in H$, one has $d \circ (i_f \circ f)(x) = d \circ (i_g \circ g)(x)$, showing $(i_f \circ f(x)) \cdot (i_g \circ g(x))^{-1} \in \ker(d)$. Hence, we have $A(S) \leq \ker(d)$ (since all of its generators are), showing that its normal closure $\overline{A(S)} \leq \ker(d)$ also. The, by first isomorphism theorem, d uniquely factors through $G * G' / \overline{A(S)}$, or there exists a unique $\bar{d} : G * G' / \overline{A(S)} \rightarrow D$, such that $d = \bar{d} \circ \pi$. Which, we have that $d_f = d \circ i_f = \bar{d} \circ (\pi \circ i_f)$, and $d_g = d \circ i_g = \bar{d} \circ (\pi \circ i_g)$, which forms the following diagram:

$$\begin{array}{ccccc}
 H & \xrightarrow{f} & G & & \\
 g \downarrow & & \downarrow \pi \circ i_f & \searrow d_f & \\
 G' & \xrightarrow{\quad} & G * G' / \overline{A(S)} & \xrightarrow{\bar{d}} & D \\
 & \searrow \pi \circ i_g & & \swarrow d_g & \\
 & & & &
 \end{array}$$

Also, the uniqueness of $\bar{d}$ can be deduced from the uniqueness of d (through the property of coproduct $G * G'$) and First Isomorphism Theorem (the property of quotient $G * G' / (\overline{A(S)})$). Hence, $G * G' / (\overline{A(S)})$ together with $\pi \circ i_f : G \rightarrow G * G' / (\overline{A(S)})$ and $\pi \circ i_g : G' \rightarrow G * G' / (\overline{A(S)})$ is indeed a fibre coproduct of f and g , showing the existence of „coproduct“ between morphism f, g , if consider them as an object in the Coslice Category $\mathbf{Grp}^H$.

3 D

Problem 4

Construct the fibre product in a category using the notation of product and equalizer. This is a hint that all small limits are built out of small products and equalizers.

Solution: Given a category $\mathcal{C}$ with product and equalizer, we'll explicitly construct small fibre product over arbitrary indexed set I .

Given Z , a collection of objects $\{X_i\}_{i \in I}$ contained in $\text{Ob}(\mathcal{C})$, and a collection of morphisms $\{f_i : X_i \rightarrow Z\}_{i \in I}$. Let $(P, \{\pi_i : P \rightarrow X_i\})$ denotes the product of the collection $\{X_i\}_{i \in I}$ (where $\pi_i : P \rightarrow X_i$ is the corresponding projection to each component). Then, the collection of morphisms $\{f_i \circ \pi_i : P \rightarrow Z\}_{i \in I}$ as an equalizer $(E, e : E \rightarrow P)$ (notice it satisfies the property that given any $i, j \in I$, such that $(f_i \circ \pi_i) \circ e = (f_j \circ \pi_j) \circ e$, and for any other pair $(A, a : A \rightarrow P)$ satisfying $(f_i \circ \pi_i) \circ a = (f_j \circ \pi_j) \circ a$ for all $i, j \in I$, there exists a unique morphism $\bar{a} : A \rightarrow E$, such that $a = e \circ \bar{a}$). Diagrammatically, we get the following for any $i, j \in I$:

$$\begin{array}{ccccc}
 E & \xrightarrow{e} & P & \begin{array}{c} \xrightarrow{f_i \circ \pi_i} \\ \xrightarrow{f_j \circ \pi_j} \end{array} & Z \\
 \uparrow \exists! \bar{a} & & \nearrow a & & \\
 A & & & &
 \end{array}$$

Now, we claim that the collection $(E, \{\pi_i \circ e : E \rightarrow X_i\}_{i \in I})$ in fact forms a fibre product for the collection of morphisms $\{f_i : X_i \rightarrow Z\}_{i \in I}$.

First, it's clear that for any $i, j \in I$, one has $f_i \circ (\pi_i \circ e) = f_j \circ (\pi_j \circ e)$ based on construction.

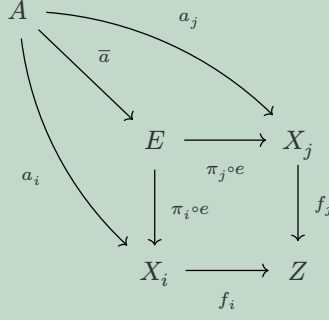
Now, given any other pair $(A, \{a_i : A \rightarrow X_i\}_{i \in I})$ that satisfies $f_i \circ a_i = f_j \circ a_j$ for all indices $i, j \in I$, notice that there exists a unique morphism $a : A \rightarrow P$ such that each $a_i = \pi_i \circ a$, based on the property of product. Diagrammatically, it's as follow for all $i, j \in I$:

$$\begin{array}{ccccc}
 & & A & & \\
 & \swarrow a_i & \vdots \exists! a & \searrow a_j & \\
 X_i & \xleftarrow{\pi_i} & P & \xrightarrow{\pi_j} & X_j
 \end{array}$$

Then, notice that since each $f_i \circ a_i = f_j \circ a_j$, with the product map it extends to $f_i \circ \pi_i \circ a = f_j \circ \pi_j \circ a$. Hence, the pair $(A, a : A \rightarrow P)$ in fact equalizes all $\{f_i \circ \pi_i : P \rightarrow Z\}_{i \in I}$, showing there must exist a unique morphism $\bar{a} : A \rightarrow E$, such that $a = e \circ \bar{a}$. diagrammatically, it is as follow:

$$\begin{array}{ccccc}
 E & \xrightarrow{e} & P & \begin{array}{c} \xrightarrow{f_i \circ \pi_i} \\ \xrightarrow{f_j \circ \pi_j} \end{array} & Z \\
 \uparrow \exists! \bar{a} & & \nearrow a & & \\
 A & & & &
 \end{array}$$

Hence, unwrap the diagram, it is presented as follow for all $i, j \in I$:



Finally, such morphism $\bar{a}$ is unique, because of $\bar{a}' : A \rightarrow E$ also satisfies such property, then notice that $e \circ \bar{a}' : A \rightarrow P$ satisfies $\pi_i \circ a = a_i = \pi_i \circ (e \circ \bar{a}')$ (if put $\bar{a}'$ in the diagram above, where $\bar{a}$ is), and since a is unique based on the property of product (when gathering the a_i s together), we must have $a = e \circ \bar{a}'$. However, we also know that $\bar{a}$ is the unique morphism satisfying $a = e \circ \bar{a}$ based on the property of equalizer, so it enforces $\bar{a}' = \bar{a}$.

Hence, given any pair $(A, \{a_i : A \rightarrow X_i\}_{i \in I})$ such that $f_i \circ a_i = f_j \circ a_j$ for all indices $i, j \in I$, one must have a unique morphism $\bar{a} : A \rightarrow E$, such that $a_i = (\pi_i \circ e) \circ \bar{a}$. This shows that $(E, \{\pi_i \circ e : E \rightarrow X_i\}_{i \in I})$ indeed satisfies the universal property of fibre product, which also proves that (small) fibre product exists if both products and equalizers exist.

4 D

Problem 5

Show that, if $F : \mathcal{C} \rightarrow \mathcal{D}$ is a natural equivalence between two categories $\mathcal{C}$ and $\mathcal{D}$, then it is both fully faithful and essentially surjective.

Solution: First, suppose F is a natural equivalence between two categories $\mathcal{C}$ and $\mathcal{D}$, then there exists another functor $G : \mathcal{D} \rightarrow \mathcal{C}$, paired with two natural isomorphisms $\eta : G \circ F \simeq \text{Id}_{\mathcal{C}}$ and $\varepsilon : F \circ G \simeq \text{Id}_{\mathcal{D}}$.

To verify it's fully faithful, given any $X, Y \in \mathcal{C}$, consider the functor's associated set map $F_{X,Y} : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$ by $F_{X,Y}(f) = Ff$, the goal is to prove that this is bijective.

For injectivity, notice that we also have $G_{F(X), F(Y)} : \text{Hom}_{\mathcal{D}}(F(X), F(Y)) \rightarrow \text{Hom}_{\mathcal{C}}(G \circ F(X), G \circ F(Y))$. Which, given any $f \in \text{Hom}_{\mathcal{C}}(X, Y)$ (where $G_{F(X), F(Y)} \circ F_{X,Y}(f) = (G \circ F)f$) satisfies the following diagram with η :

$$\begin{array}{ccc} G \circ F(X) & \xrightarrow{(G \circ F)f} & G \circ F(Y) \\ \eta_X \downarrow & & \downarrow \eta_Y \\ X & \xrightarrow{f} & Y \end{array}$$

Which, since η is a natural isomorphism, both η_X, η_Y are invertible. Hence, the map $\bar{\eta}_{X,Y} : \text{Hom}_{\mathcal{C}}(G \circ F(X), G \circ F(Y)) \rightarrow \text{Hom}_{\mathcal{C}}(X, Y)$ by $(G \circ F)f \mapsto (\eta_Y \circ (G \circ F)f \circ \eta_X) = f$ is actually an isomorphism. Combining all maps, we get that $\bar{\eta}_{X,Y} \circ G_{F(X), F(Y)} \circ F_{X,Y} : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Y)$ satisfies the following:

$$\bar{\eta}_{X,Y} \circ G_{F(X), F(Y)} \circ F_{X,Y}(f) = \bar{\eta}_{X,Y}((G \circ F)f) = f \quad (4.1)$$

Hence, this map is id on $\text{Hom}_{\mathcal{C}}(X, Y)$, showing that it's injective. So, $F_{X,Y}$ is injective.

For surjectivity instead, let $X' = F(X)$, $Y' = F(Y)$, then for any $f' \in \text{Hom}_{\mathcal{D}}(X', Y')$ (which is the codomain of $F_{X,Y}$), based on the natural isomorphism ε , we also have the following:

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ (\varepsilon_{X'})^{-1} \uparrow & & \uparrow (\varepsilon_{Y'})^{-1} \\ F \circ G(X') & \xrightarrow{(F \circ G)f'} & F \circ G(Y') \end{array}$$

Hence, define $\bar{\varepsilon}_{X', Y'} : \text{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y')) \rightarrow \text{Hom}_{\mathcal{D}}(X', Y')$ by $(F \circ G)f' \mapsto (\varepsilon_{Y'})^{-1} \circ ((F \circ G)f') \circ \varepsilon_{X'} = f'$, it is an isomorphism. So, having the maps $G_{X', Y'} : \text{Hom}_{\mathcal{D}}(X', Y') \rightarrow \text{Hom}_{\mathcal{C}}(G(X'), G(Y'))$, $F_{G(X'), G(Y')} : \text{Hom}_{\mathcal{C}}(G(X'), G(Y')) \rightarrow \text{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y'))$, the composition $\bar{\varepsilon}_{X', Y'} \circ F_{G(X'), G(Y')} \circ G_{X', Y'} : \text{Hom}_{\mathcal{D}}(X', Y') \rightarrow \text{Hom}_{\mathcal{D}}(X', Y')$ is in fact identity (since $f' \mapsto \bar{\varepsilon}_{X', Y'}((F \circ G)f') = f'$). So, with $\bar{\varepsilon}_{X', Y'}$ being an isomorphism, $F_{G(X'), G(Y')} \circ G_{X', Y'}$ is also an isomorphism, hence $F_{G(X'), G(Y')}$ is surjective.

Finally, since $\text{Hom}_{\mathcal{C}}(G(X'), G(Y')) = \text{Hom}_{\mathcal{C}}(G \circ F(X), G \circ F(Y))$ is naturally isomorphic to $\text{Hom}_{\mathcal{C}}(X, Y)$ (via η), then so is $\text{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y'))$ and $\text{Hom}_{\mathcal{D}}(F(X), F(Y))$ (via F applies to η , since functors preserve isomorphisms). Hence, if $F_{G(X'), G(Y')}$ is surjective (which has codomain $\text{Hom}_{\mathcal{D}}(F \circ G(X'), F \circ G(Y'))$), via the natural isomorphism of η , one can also conclude that $F_{X,Y}$ is also surjective (with codomain $\text{Hom}_{\mathcal{D}}(F(X), F(Y))$).

The last piece is essentially surjective: Given any $Y \in \mathcal{D}$, since we have $F \circ G(Y) \cong Y$ (due to the fact that $\varepsilon : F \circ G \simeq \text{Id}_{\mathcal{D}}$), showing it's essentially surjective.

5 ND (Split to Problem 6-12) (12 not done)

Problem 6

Definition: 1

Let R be a commutative ring. If S is any set, we denote by $L_{R(S)}$ the set of all finite linear combinations of elements in S over R – that is, the set of formal sums $\sum_i a_i \cdot s_i$ where $a_i \in R$ and $s_i \in S$, and i runs through a finite set.

Define a suitable abelian group structure on $L_{R(S)}$ and scaling of R on $L_{R(S)}$. Show that $L_{R(S)}$ is a module over R .

Solution: For this purpose, the element $0 \in L_{R(S)}$ needs to first be defined: regardless of the set $\{s_1, \dots, s_n\} \subset S$, the sum $\sum_{i=1}^n 0 \cdot s_i$ will be defined to be the same, namely the $0 \in L_{R(S)}$. Which, given any list $\{s_1, \dots, s_n\} \subset S$, if $s_{n+1} \notin \{s_1, \dots, s_n\}$, define the expression $\sum_{i=1}^n a_i \cdot s_i + 0 \cdot s_{n+1} = \sum_{i=1}^n a_i \cdot s_i$.

With the notion of 0, given any two finite formal sums $\sum_i a_i \cdot s_i, \sum_j b_j \cdot s_j$, by adding suitable terms with coefficient 0 to both expressions, WLOG one can say the two expressions are $\sum_{i=1}^n a_i \cdot s_i$ and $\sum_{i=1}^n b_i \cdot s_i$. Which, define the addition as follow:

$$\sum_{i=1}^n a_i \cdot s_i + \sum_{i=1}^n b_i \cdot s_i := \sum_{i=1}^n (a_i + b_i) \cdot s_i \quad (5.1)$$

where each $a_i + b_i \in R$ uses the addition structure of R .

Finally, for scalar multiplication, given any $r \in R$ and $\sum_i a_i \cdot s_i \in L_{R(S)}$, define scaling of r as follow:

$$r \cdot \left(\sum_i a_i \cdot s_i \right) := \sum_i (ra_i) \cdot s_i \quad (5.2)$$

To show that $L_{R(S)}$ forms a left R -Module structure, first it is evident that any $v \in L_{R(S)}$ satisfies $v + 0 = v$ (since 0 provides no extra components, and add nothing to the existing coefficients); also, addition is associative, since given $u, v, w \in L_{R(S)}$ (WLOG, say $u = \sum_{i=1}^n u_i \cdot s_i$, $v = \sum_{i=1}^n v_i \cdot s_i$, and $w = \sum_{i=1}^n w_i \cdot s_i$), then we have the following:

$$\begin{aligned} (u + v) + w &= \left(\sum_{i=1}^n u_i \cdot s_i + \sum_{i=1}^n v_i \cdot s_i \right) + \sum_{i=1}^n w_i \cdot s_i = \sum_{i=1}^n (u_i + v_i) \cdot s_i + \sum_{i=1}^n w_i \cdot s_i \\ &= \sum_{i=1}^n (u_i + v_i + w_i) \cdot s_i = \sum_{i=1}^n u_i \cdot s_i + \left(\sum_{i=1}^n v_i \cdot s_i + \sum_{i=1}^n w_i \cdot s_i \right) \\ &= u + (v + w) \end{aligned} \quad (5.3)$$

Which proves the associativity.

Now, about scalar multiplication, given any $v = \sum_i a_i \cdot s_i \in L_{R(S)}$, $0, 1 \in R$ satisfy the following:

$$0 \cdot v = 0 \cdot \left(\sum_i a_i \cdot s_i \right) = \sum_i (0 \cdot a_i) \cdot s_i = 0 \quad (5.4)$$

$$1 \cdot v = 1 \cdot \left(\sum_i a_i \cdot s_i \right) = \sum_i (1 \cdot a_i) \cdot s_i = v \quad (5.5)$$

Finally, about distributivity, given any $r, t \in R$ and $u, v \in L_{R(S)}$ (say $u = \sum_i u_i \cdot s_i$ and $v = \sum_i v_i \cdot s_i$), the following two equations are true:

$$\begin{aligned} (r+t) \cdot (u+v) &= (r+t) \cdot \left(\sum_i u_i \cdot s_i + \sum_i v_i \cdot s_i \right) \\ &= (r+t) \cdot \sum_i (u_i + v_i) \cdot s_i = \sum_i (ru_i + rv_i + tu_i + tv_i) \cdot s_i \\ &= \sum_i (ru_i) \cdot s_i + \sum_i (rv_i) \cdot s_i + \sum_i (tu_i) \cdot s_i + \sum_i (tv_i) \cdot s_i \\ &= r \cdot \left(\sum_i u_i \cdot s_i \right) + r \cdot \left(\sum_i v_i \cdot s_i \right) + t \cdot \left(\sum_i u_i \cdot s_i \right) + t \cdot \left(\sum_i v_i \cdot s_i \right) \\ &= r \cdot u + r \cdot v + t \cdot u + t \cdot v \end{aligned} \quad (5.6)$$

$$\begin{aligned} (rt) \cdot v &= (rt) \cdot \left(\sum_i u_i \cdot s_i \right) = \sum_{i(rt u_i)} \cdot s_i = r \cdot \left(\sum_i (tu_i) \cdot s_i \right) \\ &= r \cdot \left(t \cdot \left(\sum_i u_i \cdot s_i \right) \right) = r \cdot (t \cdot v) \end{aligned} \quad (5.7)$$

Hence, $L_{R(S)}$ under such addition / multiplication structure forms a left R -module.

Problem 7

Suppose that $R = \mathbb{R}$ is the ring/field of real numbers and let $S = \mathbb{N}$ denote the set of natural numbers, what is the $\mathbb{R}$ -vector space $L_{\mathbb{R}}(\mathbb{N})$? Can you find a better way to write this vector space?

Solution: The vector space $L_{\mathbb{R}}(\mathbb{N})$ can also be characterized as the direct sum $\bigoplus_{n \in \mathbb{N}} \mathbb{R}$ (since there are countable generators, defined with finite sums, and in between any two generators there is no relations).

Another way of characterizing it can be thought as follow: Consider the set of functions $F = \{f : \mathbb{N} \rightarrow \mathbb{R} \mid f(n) \neq 0 \text{ for finite } n \in \mathbb{N}\}$, which because the codomain is $\mathbb{R}$ (which has a natural addition / scalar multiplication structure on itself) then this automatically forms F into an $\mathbb{R}$ -vector space.

Now, for each $n \in \mathbb{N}$, let $e_n : \mathbb{N} \rightarrow \mathbb{R}$ denotes the function satisfying $e_n(m) = \delta_{nm}$ (the kronecker delta). Then, notice that for each function $f \in F$, if $n_1, \dots, n_k \in \mathbb{N}$ are the only natural numbers being evaluated to have nonzero output, then $f = \sum_{i=1}^k f(n_i) \cdot e_{n_i}$:

$$m \notin \{n_1, \dots, n_k\} \implies \left(\sum_{i=1}^k f(n_i) \cdot e_{n_i} \right)(m) = 0 \quad (5.8)$$

$$\forall j \in \{1, \dots, k\}, \quad \left(\sum_{i=1}^k f(n_i) \cdot e_{n_i} \right)(n_j) = \sum_{i=1}^k f(n_i) \cdot \delta_{n_i n_j} = f(n_j) \cdot 1 = f(n_j) \quad (5.9)$$

Hence, $f = \sum_{i=1}^k f(n_i) \cdot e_{n_i}$, showing the list $\{e_n\}_{n \in \mathbb{N}}$ in fact generates F .

Also, it is evident that each e_n has no relation with each other (which can be shown through linear independence): Given any list $e_{n_1}, \dots, e_{n_k}$, suppose $a_1, \dots, a_k \in \mathbb{R}$ satisfies $\sum_{i=1}^k a_i \cdot e_{n_i} = 0$, then for each $j \in \{1, \dots, k\}$, the following is true:

$$0 = \left(\sum_{i=1}^k a_i \cdot e_{n_i} \right)(n_j) = \sum_{i=1}^k a_i \cdot \delta_{n_i n_j} = a_j \cdot 1 = a_j \quad (5.10)$$

Hence, each $a_j = 0$, showing that the list is linearly independent. This shows that $\{e_n\}_{n \in \mathbb{N}}$ forms a basis of the vector space F (which they're linearly independent, and their finite $\mathbb{R}$ -linear combinations generate the whole F). So, this is another way of identifying $L_{\mathbb{R}}(\mathbb{N})$.

Problem 8**Definition: 2**

Consider the module $L_R(M \times N)$ and define a submodule $L' \subset L_R(M \times N)$ generated by all elements of the following form:

- $r(x, y) - (rx, y)$
- $r(x, y) - (x, ry)$
- $(x_1 + x_2, y) - ((x_1, y) + (x_2, y))$
- $(x, y_1 + y_2) - ((x, y_1) + (x, y_2))$

where $r \in R$, $x, x_1, x_2 \in M$, and $y, y_1, y_2 \in N$. We then define the quotient module

$$M \otimes_R N := L_R(M \times N)/L' \quad (5.11)$$

and call it the tensor product of M and N .

Definition: 3

If P is another R -module, a bilinear map is a map $B : M \times N \rightarrow P$ that is linear with respect to both entries – that is, for all $x_0 \in M$ the map $B(x_0, _) : N \rightarrow P$ is linear and for all $y_0 \in N$ the map $B(_, y_0) : M \rightarrow P$ is linear.

Construct a natural bilinear map $\pi : M \times N \rightarrow M \otimes_R N$ and show that it is universal in the following sense: for any bilinear map $B : M \times N \rightarrow P$, there is a unique *linear* map $h : M \otimes_R N \rightarrow P$ such that $B = h \circ \pi$. Conclude that for any R -module P , we have a canonical identification

$$\text{Hom}_R(M \otimes_R N, P) \cong \text{Bil}(M \times N, P) \quad (5.12)$$

between the set of R -module homomorphisms from $M \otimes_R N$ to P , and the set of bilinear maps from $M \times N$ to P .

Solution:**I. The Bilinear map π :**

For simplicity, given any $m \in M$ and $n \in N$, we'll denote $m \otimes n := \overline{(m, n)} \in M \otimes_R N = L_R(M \times N)/L'$ (the quotient of (m, n) in $M \otimes_R N$). Then, given the relation of L' , we get the following relation:

- $r(x \otimes y) = (rx) \otimes y$, and $r(x \otimes y) = x \otimes (ry)$.
- $(x_1 + x_2) \otimes y = x_1 \otimes y + x_2 \otimes y$.
- $x \otimes (y_1 + y_2) = x \otimes y_1 + x \otimes y_2$.

Where $r \in R$, $x, x_1, x_2 \in M$ and $y, y_1, y_2 \in N$.

Hence, define the map $\pi : M \times N \rightarrow M \otimes_R N$ by $\pi(m, n) = m \otimes n$, this map is well-defined (since each $(m, n) \in M \times N$ corresponds to a unique representative in $M \otimes_R N := L_R(M \times N)/L'$). Then, by the property of tensor products, one gets the following for any $r, t \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$:

$$\begin{aligned} \pi(rm_1 + tm_2, n) &= (rm_1 + tm_2) \otimes n = (rm_1) \otimes n + (tm_2) \otimes n \\ &= r(m_1 \otimes n) + t(m_2 \otimes n) = r \cdot \pi(m_1, n) + t \cdot \pi(m_2, n) \end{aligned} \quad (5.13)$$

$$\begin{aligned} \pi(m, rn_1 + tn_2) &= m \otimes (rn_1 + tn_2) = m \otimes (rn_1) + m \otimes (tn_2) \\ &= r(m \otimes n_1) + t(m \otimes n_2) = r \cdot \pi(m, n_1) + t \cdot \pi(m, n_2) \end{aligned} \quad (5.14)$$

The above two equations show that π is naturally bilinear.

II. Universal Property of π :

Given any R -module P together with a bilinear map $B : M \times N \rightarrow P$, suppose for the moment that there exists an R -linear map $h : M \otimes_R N \rightarrow P$ satisfies $B = h \circ \pi$, then for all $m \in M$ and $n \in N$, we have the following:

$$B(m, n) = h \circ \pi(m, n) = h(m \otimes n) \quad (5.15)$$

Now, notice that since $L_R(M \times N)$ is generated by finite formal linear combination of $(m, n) \in M \times N$, then $M \otimes_R N = L_{R(M \times N)} / L'$ is instead generated by finite linear combination of the quotient $\overline{(m, n)} = m \otimes n$. Then, since h above must satisfy $h(m \otimes n) = B(m, n)$, then h fixes where the generators of $M \otimes_R N$ goes, hence fixes the whole R -linear map h . This implies that h is uniquely determined by B .

Then, to show that h exists, it suffices to show that it's well-defined on the generators (since the definition of being an R -linear map automatically extends it to be well-defined on any finite R -linear combinations of $m \otimes n \in M \otimes_R N$). To show that any possible representatives eventually gets send to the same element via h , consider any $a, b, c, d \in R$, $m_1, m_2 \in M$, and $n_1, n_2 \in N$. Which, the following is true:

$$\begin{aligned} B(am_1 + bm_2, cn_1 + dn_2) &= (ac) \cdot B(m_1, n_1) + (ad) \cdot B(m_1, n_2) \\ &\quad + (bc) \cdot B(m_2, n_1) + (bd) \cdot B(m_2, n_2) \end{aligned} \quad (5.16)$$

The above is based on the bilinearity of B . Which, notice that on the tensor product level, the following is true:

$$\begin{aligned} (am_1 + bm_2) \otimes (cn_1 + dn_2) &= (ac) \cdot (m_1 \otimes n_1) + (ad) \cdot (m_1 \otimes n_2) \\ &\quad + (bc) \cdot (m_2 \otimes n_1) + (bd) \cdot (m_2 \otimes n_2) \end{aligned} \quad (5.17)$$

Hence, this shows the following:

$$\begin{aligned} h((am_1 + bm_2) \otimes (cn_1 + dn_2)) &= B(am_1 + bm_2, cn_1 + dn_2) \\ &= (ac) \cdot B(m_1, n_1) + (ad) \cdot B(m_1, n_2) \\ &\quad + (bc) \cdot B(m_2, n_1) + (bd) \cdot B(m_2, n_2) \\ &= (ac) \cdot h(m_1 \otimes n_1) + (ad) \cdot h(m_1 \otimes n_2) \\ &\quad + (bc) \cdot h(m_2 \otimes n_1) + (bd) \cdot h(m_2 \otimes n_2) \\ &= h((ac) \cdot (m_1 \otimes n_1) + (ad) \cdot (m_1 \otimes n_2) \\ &\quad + (bc) \cdot (m_2 \otimes n_1) + (bd) \cdot (m_2 \otimes n_2)) \end{aligned} \quad (5.18)$$

Which, it implies that the definition $h(m \otimes n) = B(m, n)$ (together with h being an R -linear map) is well-defined (since under different representatives of the same element, h still acts the same). Hence, as an R -linear map $h : M \otimes_R N \rightarrow P$ by $h(m \otimes n) = B(m, n)$ is well-defined.

This finishes the proof about the universal property, that every bilinear map $B : M \times N \rightarrow P$ admits a unique R -linear map $h : M \otimes_R N \rightarrow P$ with $B = h \circ \pi$.

III. Isomorphism between $\text{Hom}_R(M \otimes_R N, P)$ and $\text{Bil}(M \times N, P)$:

First, notice that linear combinations of R -linear maps (respectively, R -Bilinear maps) are still R -linear map (respectively, R -Bilinear map). Hence, $\text{Hom}_R(M \otimes_R N, P)$, $\text{Bil}(M \times N, P)$ are both R -modules also.

Now, notice that the map $_ \circ \pi : \text{Hom}_R(M \otimes_R N, P) \rightarrow \text{Bil}(M \times N, P)$ is a module homomorphism:

- First, given any R -linear map $h : M \otimes_R N \rightarrow P$, we have the following for any $r, t \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$:

$$\begin{aligned} h \circ \pi(rm_1 + tm_2, n) &= h((rm_1 + tm_2) \otimes n) = h(r(m_1 \otimes n) + t(m_2 \otimes n)) \\ &= r \cdot h(m_1 \otimes n) + t \cdot h(m_2 \otimes n) \\ &= r \cdot h \circ \pi(m_1, n) + t \cdot h \circ \pi(m_2, n) \end{aligned} \quad (5.19)$$

$$\begin{aligned} h \circ \pi(m, rn_1 + tn_2) &= h(m \otimes (rn_1 + tn_2)) = h(r(m \otimes n_1) + t(m \otimes n_2)) \\ &= r \cdot h(m \otimes n_1) + t \cdot h(m \otimes n_2) \\ &= r \cdot h \circ \pi(m, n_1) + t \cdot h \circ \pi(m, n_2) \end{aligned} \quad (5.20)$$

The two equalities show that $h \circ \pi$ is a bilinear map, hence $h \circ \pi \in \text{Bil}(M \times N, P)$, the map $_ \circ \pi$ is well-defined.

The reason it's an R -linear map, is because given any $f, g \in \text{Hom}_R(M \otimes_R N, P)$, $a, b \in R$, and any $m \in M$ and $n \in N$, the following is true:

$$\begin{aligned} (a \cdot f + b \cdot g) \circ \pi(m, n) &= (a \cdot f + b \cdot g)(m \otimes n) \\ &= a \cdot f(m \otimes n) + b \cdot g(m \otimes n) \\ &= a \cdot f \circ \pi(m, n) + b \cdot g \circ \pi(m, n) \end{aligned} \quad (5.21)$$

Hence, $(a \cdot f + b \cdot g) \circ \pi = a \cdot (f \circ \pi) + b \cdot (g \circ \pi)$, showing $_ \circ \pi$ is an R -linear map.

Finally, it's a bijection, because for any bilinear map $B \in \text{Bil}(M \times N, P)$, the universal property states there exists $h \in \text{Hom}_R(M \otimes_R N, P)$ such that $B = h \circ \pi$, showing $_ \circ \pi$ is surjective; and, it's injective is because if $h, h' \in \text{Hom}_R(M \otimes_R N, P)$ satisfies $B = h \circ \pi = h' \circ \pi \in \text{Bil}(M \times N, P)$, since both h, h' satisfies $B = h \circ \pi = h' \circ \pi$, the universal property again states $h = h'$, hence $_ \circ \pi$ is injective.

So, this identifies that $\text{Hom}_{R(M \otimes_R N, P)} \cong \text{Bil}(M \times N, P)$.

Problem 9

Is $M \otimes_R N$ the categorical pushout of M and N in the category **R-Mod** of R -modules? If so, prove it. If not, what is the pushout of M and N ?

Solution: We'll first construct the pushout (or coproduct) of any R -module M and N , and show that in general it's different from $M \otimes_R N$ by providing a counterexample.

I. Coproduct in R-Mod:

Given arbitrary R -modules M, N , consider the direct sum $M \oplus N$ together with the two canonical injection $\iota_1 : M \rightarrow M \oplus N$ and $\iota_2 : N \rightarrow M \oplus N$. It is evident that it satisfies the first property of coproduct (which are two maps from M, N respectively into the R -module).

Then, to show its universality, given any other R -module P together with two maps $h_1 : M \rightarrow P$ and $h_2 : N \rightarrow P$, the map $h : M \oplus N \rightarrow P$ by $h(m, n) = h_1(m) + h_2(n)$ satisfies the following:

- First, given any $(m, n), (m', n') \in M \oplus N$, and $a, b \in R$, the following is true:

$$\begin{aligned} h(a(m, n) + b(m', n')) &= h(am + bm', an + bn') = h_1(am + bm') + h_2(an + bn') \\ &= ah_1(m) + bh_1(m') + ah_2(n) + bh_2(n') \\ &= a(h_1(m) + h_2(n)) + b(h_1(m') + h_2(n')) \\ &= a \cdot h(m, n) + b \cdot h(m', n') \end{aligned} \quad (5.22)$$

Hence, h is a Module homomorphism.

- For any $m \in M$ and $n \in N$, we have the following two equality:

$$h \circ \iota_1(m) = h(m, 0) = h_1(m) + h_2(0) = h_1(m) \quad (5.23)$$

$$h \circ \iota_2(n) = h(0, n) = h_1(0) + h_2(n) = h_2(n) \quad (5.24)$$

Hence, $h \circ \iota_1 = h_1$ and $h \circ \iota_2 = h_2$.

- h is the unique morphism with such property: Suppose given $h' : M \oplus N \rightarrow P$ also satisfies such property, then we must have $h' \circ \iota_1 = h_1$ and $h' \circ \iota_2 = h_2$. Then, for all $m \in M$ and $n \in N$, we must have $h_1(m) = h' \circ \iota_1(m) = h'(m, 0)$, and $h_2(n) = h' \circ \iota_2(n) = h'(0, n)$. So, $h'(m, n) = h'(m, 0) + h'(0, n) = h_1(m) + h_2(n)$, showing that $h' = h$.

These conditions show that $M \oplus N$ together with the two canonical injections $\iota_1 : M \rightarrow M \oplus N$ and $\iota_2 : N \rightarrow M \oplus N$, is initial with such property. Meaning given the pair $(P, h_1 : M \rightarrow P, h_2 : N \rightarrow P)$, there exists unique $h : M \oplus N \rightarrow P$, such that the following diagram commutes:

$$\begin{array}{ccccc} M & \xrightarrow{\iota_1} & M \oplus N & \xleftarrow{\iota_2} & N \\ & \searrow h_1 & \downarrow \exists! h & \swarrow h_2 & \\ & & P & & \end{array}$$

Hence, $M \oplus N$ with canonical injection ι_1, ι_2 forms the coproduct of M, N in **R-Mod**.

II. Counterexample to $M \otimes_R N$ being Pushout/Coproduct:

Here's a simple one: Consider $0 \otimes_{\mathbb{Z}} \mathbb{Z}$, since everything is in the form $0 \otimes n$ for some $n \in \mathbb{Z}$; then by the property of tensor product, $2 \cdot (0 \otimes n) = (n \cdot 0) \otimes n = 0 \otimes n$, hence $0 \otimes n = 0$, or $0 \otimes_{\mathbb{Z}} \mathbb{Z} = 0$. However, the actual coproduct is $0 \oplus \mathbb{Z} \cong \mathbb{Z} \neq 0$, showing that $0 \otimes_{\mathbb{Z}} \mathbb{Z}$ is actually not a coproduct in **$\mathbb{Z}$ -Mod**.

Problem 10

Given A, B two commutative rings with ring homomorphisms $R \rightarrow A$ and $R \rightarrow B$, this makes A, B into R -modules. Let $A \otimes_R B$ denote the tensor product of A and B as R -modules.

In this setup, show that $A \otimes_R B$ is again a (commutative) R -algebra.

Solution: Here, assume we know that tensor product (of multiple R -modules) is associative (up to isomorphism), and the property that any multilinear map uniquely factors through tensor products. Then, since A, B both have their own ring structure, then consider the following R -multilinear map $Q : A \times B \times A \times B \rightarrow A \otimes_R B$:

$$Q(a, b, a', b') = (aa') \otimes (bb') \quad (5.25)$$

This is multilinear, because both ring multiplication and tensor products are bilinear. Then, the property of tensor product implies that it uniquely factors through $(A \otimes_R B) \otimes_R (A \otimes_R B)$, say through an R -linear map $\bar{Q} : (A \otimes_R B) \otimes_R (A \otimes_R B) \rightarrow A \otimes_R B$ as follows:

$$\bar{Q}((a \otimes b) \otimes (a' \otimes b')) = (aa') \otimes (bb') \quad (5.26)$$

Also, if we consider the bilinear projection $\pi : (A \otimes_R B) \times (A \otimes_R B) \rightarrow (A \otimes_R B) \otimes_R (A \otimes_R B)$ by $\pi(a \otimes b, a' \otimes b') = (a \otimes b) \otimes (a' \otimes b')$, then the bilinear map $M = \bar{Q} \circ \pi : (A \otimes_R B) \times (A \otimes_R B) \rightarrow A \otimes_R B$ as follows:

$$M(a \otimes b, a' \otimes b') = \bar{Q}((a \otimes b) \otimes (a' \otimes b')) = (aa') \otimes (bb') \quad (5.27)$$

Which, M (as an R -bilinear map) can also be viewed as a multiplication on $A \otimes_R B$, since it satisfies associativity as follows:

$$\begin{aligned} M(a \otimes b, M(a' \otimes b', a'' \otimes b'')) &= M(a \otimes b, (a'a'') \otimes (b'b'')) = (a(a'a'')) \otimes (b(b'b'')) \\ &= ((aa')a'') \otimes ((bb')b'') = M((aa') \otimes (bb'), a'' \otimes b'') \quad (5.28) \\ &= M(M(a \otimes b, a' \otimes b'), a'' \otimes b'') \end{aligned}$$

Also, distributivity naturally follows from the fact that M is bilinear. Hence, this makes $A \otimes_R B$ into an R -algebra, with multiplication M .

Problem 11

Prove that $A \otimes_R B$ is the pushout (coproduct) of A and B in the category $\mathbf{R}\text{-Alg}$ of R -algebras.

Solution: First, it's clear that given $A \otimes_R B$, there are natural R -algebra homomorphisms $i_1 : A \rightarrow A \otimes_R B$ and $i_2 : B \rightarrow A \otimes_R B$ by $i_1(a) = a \otimes 1$ and $i_2(b) = 1 \otimes b$ (which since $i_1(r \cdot a) = (m_A(r)a) \otimes 1 = r \cdot (a \otimes 1) = r \cdot i_1(a)$, it's an R -module homomorphism together with the additive property; also, we have $i_1(aa') = (aa') \otimes 1 = (a \otimes 1)(a' \otimes 1) = i_1(a) \cdot i_1(a')$, which is also an R -algebra homomorphism. Similar logic applies to i_2).

Now, given any other R -algebra C , and suppose two R -algebra homomorphisms $f : A \rightarrow C$ and $g : B \rightarrow C$ are given. Then, if we consider the map $B : A \times B \rightarrow C$ by $B(a, b) = f(a)g(b)$, notice that this map is R -bilinear. Given any $r, t \in R$, $a, a' \in A$, and $b, b' \in B$, we have the following:

$$\begin{aligned} B(r \cdot a + t \cdot a', b) &= f(r \cdot a + t \cdot a')g(b) = (r \cdot f(a) + t \cdot f(a'))g(b) \\ &= r \cdot (f(a)g(b)) + t \cdot (f(a')g(b)) = r \cdot B(a, b) + t \cdot B(a', b) \end{aligned} \quad (5.29)$$

$$\begin{aligned} B(a, r \cdot b + t \cdot b') &= f(a)g(r \cdot b + t \cdot b') = f(a)(r \cdot g(b) + t \cdot g(b')) \\ &= r \cdot (f(a)g(b)) + t \cdot (f(a)g(b')) = r \cdot B(a, b) + t \cdot B(a, b') \end{aligned} \quad (5.30)$$

This shows that B is R -bilinear, hence it uniquely factors through the tensor product, say by the R -module homomorphism $h : A \otimes_R B \rightarrow C$ by $h(a \otimes b) = f(a)g(b)$.

However, this also turns h into a ring homomorphism, since we have the following:

$$\begin{aligned} h((a \otimes b) \cdot (a' \otimes b')) &= h((aa') \otimes (bb')) = f(aa')g(bb') = f(a)f(a')g(b)g(b') \\ &= (f(a)g(b)) \cdot (f(a')g(b')) = h(a \otimes b) \cdot h(a' \otimes b') \end{aligned} \quad (5.31)$$

Since it's both an R -module homomorphism and a commutative ring homomorphism, h is in fact an R -algebra homomorphism. And, it satisfies the following, for all $a \in A$ and $b \in B$:

$$f(a) = f(a)g(1) = h(a \otimes 1) = h \circ i_1(a) \quad (5.32)$$

$$g(b) = f(1)g(b) = h(1 \otimes b) = h \circ i_2(b) \quad (5.33)$$

Hence, $f = h \circ i_1$ and $g = h \circ i_2$.

Finally, to show uniqueness of such map h , suppose some R -algebra homomorphism $h' : A \otimes_R B \rightarrow C$ satisfies $f = h' \circ i_1$ and $g = h' \circ i_2$ also, then for all $a \in A$ and $b \in B$, we have the following:

$$\begin{aligned} h'(a \otimes b) &= h'((a \otimes 1)(1 \otimes b)) = h'(a \otimes 1) \cdot h'(1 \otimes b) \\ &= (h' \circ i_1(a)) \cdot (h' \circ i_2(b)) = f(a)g(b) = h(a \otimes b) \end{aligned} \quad (5.34)$$

This shows that $h' = h$, proving the uniqueness. Hence, $A \otimes_R B$, $i_1 : A \rightarrow A \otimes_R B$, and $i_2 : B \rightarrow A \otimes_R B$ satisfies the following commutative diagram (given some other R -algebra C with $f : A \rightarrow C$ and $g : B \rightarrow C$ being R -algebra homomorphisms):

$$\begin{array}{ccccc}
 A & \xrightarrow{i_1} & A \otimes_R B & \xleftarrow{i_2} & B \\
 & \searrow f & \downarrow \exists! h & \swarrow g & \\
 & & C & &
 \end{array}$$

So, $A \otimes_R B$ serves as a coproduct in $\mathbf{R}\text{-Alg}$, category of commutative R -algebra.

Problem 12

Compute the following tensor products:

- $k[x] \otimes_k k[x]$.
- $k[x] \otimes_{k[x]} k[x]$ where both $k[x]$ are viewed as $k[x]$ -algebras via the identity map.
- $k[x] \otimes_{k[x]} k[y]$, where $k[x]$ is a $k[x]$ -algebra via the identity map and $k[y]$ is viewed as a $k[x]$ -algebra via the map $k[x] \rightarrow k[y]$ sending $x \mapsto y^2$.
- $k[y] \otimes_{k[x]} k[y]$, where $k[y]$ is viewed as a $k[x]$ -algebra via the map $k[x] \rightarrow k[y]$ sending $x \mapsto y^2$.

Solution:

1. For any k -algebra A , given any two k -algebra homomorphism $f, g : k[x] \rightarrow A$, since one must have $f(l) = l \cdot f(1) = l \cdot 1 \in A$ (so is $g(l) = l \cdot g(1) = l \cdot 1 \in A$) for all $l \in k$. Hence, knowing $f(x)$ and $g(x)$ uniquely determines the whole map f, g .

For simplicity, the second map we'll use y instead of x , or $g : k[y] \rightarrow A$ to prevent confusion. Then, notice that given the polynomial ring $k[x, y]$, f and g uniquely determines a ring homomorphism $h : k[x, y] \rightarrow A$ by $h(x) = f(x)$ and $h(y) = g(y)$. Such map is well-defined, since A is commutative, and the fact that $f(l) = g(l)$ for all $l \in k$ (so for everything inside k , where $k[x]$ and $k[y]$ intersects, h is a well-defined map). Also, h is a ring homomorphism because A is commutative, while f, g are both ring homomorphism (so act on individual components of any polynomial in $k[x, y]$ makes sense).

Now, if consider the canonical inclusion $\iota_x : k[x] \hookrightarrow k[x, y]$ and $\iota_y : k[y] \hookrightarrow k[x, y]$, one has the following:

$$f(p(x)) = f(p(x)) \cdot g(1) = h(p(x) \cdot 1) = h(\iota_x(p(x))) \quad (5.35)$$

$$g(q(y)) = f(1) \cdot g(q(y)) = h(1 \cdot q(y)) = h(\iota_y(q(y))) \quad (5.36)$$

Hence, $f = h \circ \iota_x$ and $g = h \circ \iota_y$, so the following diagram commutes:

$$\begin{array}{ccccc} k[x] & \xrightarrow{\iota_x} & k[x, y] & \xleftarrow{\iota_y} & k[y] \\ & \searrow f & \downarrow h & \swarrow g & \\ & & A & & \end{array}$$

Also, notice the map h is unique, since if $h' : k[x, y] \rightarrow A$ as a k -algebra homomorphism also satisfies the above properties, then $h'(x) = h' \circ \iota_x(x) = f(x)$ and $h'(y) = h' \circ \iota_y(y) = g(y)$. This shows that $h'(x) = h(x)$ and $h'(y) = h(y)$, hence $h' = h$ (since they're the same on the generators, while must agree on k due to the property of k -algebra homomorphism). Hence, $k[x, y]$ with ι_x, ι_y is indeed a coproduct of $k[x], k[y]$ in the category of commutative k -algebra, showing that $k[x, y] \cong k[x] \otimes_k k[y]$ (here, if view y as x again, it's $k[x] \otimes_k k[x]$).

2. For this, we'll prove a following lemma:

Lemma

Given any commutative ring R , any R -module M satisfies $M \cong R \otimes_R M$. This in particular is true for commutative R -algebra also.

Proof: For any element in $R \otimes_R M$ is a finite linear combination of $r \otimes m$, where $r \in R$ and $m \in M$. Using the property of tensor product, $r \otimes m = 1 \otimes (r \cdot m)$.

Hence, define the R -module homomorphism $\varphi : M \rightarrow R \otimes_R M$ by $\varphi(m) = 1 \otimes m$ (can also be thought of as the bilinear projection map $\pi : R \times M \rightarrow R \otimes_R M$ by restricting the first entry to be 1). On the other hand, if consider the scalar multiplication map $s : R \times M \rightarrow M$ by $s(r, m) = r \cdot m$, s is also a bilinear map, hence it uniquely factors through the tensor product, say having R -module homomorphism $\bar{s} : R \otimes_R M \rightarrow M$ by $s = \bar{s} \circ \pi$ (so we have $\bar{s}(r \otimes m) = \bar{s} \circ \pi(r, m) = s(r, m) = r \cdot m$).

Notice that φ and $\bar{s}$ are mutual inverse: Given any $r \in R$ and $m \in M$, the following are true:

$$\bar{s} \circ \varphi(m) = \bar{s}(1 \otimes m) = s(1, m) = 1 \cdot m = m \quad (5.37)$$

$$\varphi \circ \bar{s}(r \otimes m) = \varphi(r \cdot m) = 1 \otimes (r \cdot m) = r \otimes m \quad (5.38)$$

Hence, this shows that $M \cong R \otimes_R M$ as R -modules. $\square$

As a result of the lemma, since $k[x]$ is a $k[x]$ -module (with natural left multiplication on itself), then $k[x] \otimes_{k[x]} k[x] \cong k[x]$ as $k[x]$ -modules (via the map $\bar{s} : k[x] \otimes_{k[x]} k[x] \rightarrow k[x]$ that's defined as $f(x) \otimes g(x) \mapsto f(x) \cdot g(x)$).

However, notice that this is also multiplicative, since the following is true:

$$\begin{aligned} \bar{s}((f(x) \otimes g(x)) \cdot (h(x) \otimes k(x))) &= \bar{s}((f \cdot h)(x) \otimes (g \cdot k)(x)) \\ &= (f \cdot h \cdot g \cdot k)(x) = (f \cdot g)(x) \cdot (h \cdot k)(x) \\ &= \bar{s}(f(x) \otimes g(x)) \cdot \bar{s}(h(x) \otimes k(x)) \end{aligned} \quad (5.39)$$

Hence, the two are actually isomorphic as rings, which is isomorphic as $k[x]$ -algebras.

So, $k[x] \otimes_{k[x]} k[x] \cong k[x]$.

3. Similar in this case, using the previous lemma we know $k[x] \otimes_{k[x]} k[y] \cong k[y]$ via the map $\bar{s}(f(x) \otimes g(y)) = f(x) \cdot g(y) = f(y^2)g(y)$ (Note: Since $k[y]$ is now a $k[x]$ -algebra via the map $x \mapsto y^2$, so $f(x) \cdot g(y) = f(y^2)g(y)$). This is also true on the level of $k[x]$ -algebra homomorphism, since the following is true:

$$\begin{aligned} \bar{s}((f(x) \otimes g(y)) \cdot (h(x) \otimes k(y))) &= \bar{s}((f \cdot h)(x) \otimes (g \cdot k)(y)) \\ &= (f \cdot h)(y^2)(g \cdot k)(y) = (f(y^2)g(y)) \cdot (h(y^2)k(y)) \\ &= \bar{s}(f(x) \otimes g(y)) \cdot \bar{s}(h(x) \otimes k(y)) \end{aligned} \quad (5.40)$$

So, $k[x] \otimes_{k[x]} k[y] \cong k[y]$ as $k[x]$ -algebra also.

Now, notice that if we consider the ring homomorphism $\varphi : k[x, y] \rightarrow k[y]$ by $\varphi(f(x, y)) = f(y^2, y)$, this is surjective (since for any $f(y) \in k[y] \subset k[x, y]$, it has $\varphi(f) = f(y)$), so $k[y] \cong k[x, y] / \ker(\varphi)$. Which, given the polynomial $y^2 - x$, we have $\varphi(y^2 - x) = y^2 - y^2 = 0$, so $y^2 - x \in \ker(\varphi)$.

Also, if consider any $f(x, y) \in \ker(\varphi)$ (i.e. $f(y^2, y) = 0$), to prove that $f(x, y) \in (y^2 - x)$, we'll induct on the degree of x :

- For $\deg_x(f) = 1$, since $f(x, y) = f_1(y) \cdot x + f_0(y)$ for some $f_1, f_0 \in k[y]$, then $0 = \varphi(f) = f_1(y) \cdot y^2 + f_0(y)$. If $f_1(y) \neq 0$, then it has degree at least 2, so in case for the whole thing to be 0, $f_0(y)$ cannot have degree 0 and degree 1 terms (since these two terms only appear in $f_0(y)$).

Hence, $f_0(y) = -y^2 \cdot f_1(y)$ for some $f_1 \in k[y]$. So, $f_1(y) \cdot y^2 + f_0(y) = f_1(y) \cdot y^2 - y^2 \cdot f_1(y) = 0$, showing that $y^2(f_1(y) + f_0(y)) = 0$. With $k[y]$ being an integral domain, $f_1(y) + f_0(y) = 0$, so $f_1(y) = -f_0(y)$, showing the following:

$$f(x, y) = f_1(y) \cdot x + f_0(y) = -g_0(y) \cdot x + g_0(y) \cdot y^2 = g_0(y) \cdot (y^2 - x) \quad (5.41)$$

Hence, $f(x, y) \in (y^2 - x)$.

- Now, suppose for degree $\deg_x(f) = k < n$, we always have $f(x, y) \in \ker(\varphi) \implies f(x, y) \in (y^2 - x)$. Then, if $\deg_x(f) = n$, we again decompose $f(x, y) = f_1(x, y) \cdot x + f_0(y)$, where $\deg_x(f_1) = n - 1 < n$. Then, do the following modification, we get:

$$f(x, y) = f_1(x, y) \cdot x + f_0(y) = f_1(x, y) \cdot (x - y^2) + f_1(x, y) \cdot y^2 + f_0(y) \quad (5.42)$$

Now, it's evident that $f_1(x, y) \cdot (x - y^2) \in (y^2 - x)$ (since it has a factor); also, if consider $g(x, y) = f_1(x, y) \cdot y^2 + f_0(y)$, since $f_1(x, y)$ is the only component including x , then $\deg_x(h) = \deg_x(f_1) = n - 1 < n$. Then, since $h \in \ker(\varphi)$ (since $h(x, y) = f(x, y) - f_1(x, y)(x - y^2)$, which are both in $\ker(\varphi)$), and $\deg_x(h) < n$, by induction hypothesis one must have $h \in (y^2 - x)$ also.

Hence, $f(x, y) = f_1(x, y) \cdot (x - y^2) + h(x, y) \in (y^2 - x)$. This completes the induction.

So, eventually we get that $f(x, y) \in \ker(\varphi) \implies f(x, y) \in (y^2 - x)$, showing $\ker(\varphi) \subseteq (y^2 - x)$. Hence, $\ker(\varphi) = (y^2 - x)$, which we concluded the following:

$$k[x] \otimes_{k[x]} k[y] \cong k[y] \cong k[x, y] / \ker(\varphi) = k[x, y] / (y^2 - x) \quad (5.43)$$

4. For $k[y] \otimes_{k[x]} k[z]$ (where $k[y]$ is given in the previous part), for less confusion the second one we'll denote as $k[z]$ as $k[x]$ -algebra (by the map $k[x] \rightarrow k[z]$ through $x \mapsto z^2$). Since $k[x, y] := (k[x])[y]$ satisfies $(k[x])[y] \otimes_{k[x]} (k[x])[z] \cong (k[x])[y, z]$ (based on part 1), in $k[x]$ -algebra, we have the coproduct of $k[x, y]$ and $k[x, z]$ being $k[x, y, z]$.

Now, based on part 3, we know that $k[y] \cong k[x, y] / (y^2 - x)$ and $k[z] \cong k[x, z] / (z^2 - x)$. So, based on the property of tensor product (as coproduct in $k[x]$ -algebra), with the following maps:

- inclusion map $i_1 : k[x, y] \hookrightarrow k[x, y, z]$, $i_2 : k[x, z] \hookrightarrow k[x, y, z]$;
- quotient map $\pi_1 : k[x, y] \twoheadrightarrow k[x, y] / (y^2 - x)$, $\pi_2 : k[x, z] \twoheadrightarrow k[x, z] / (z^2 - x)$;
- injection to tensor product $\iota_1 : k[x, y] / (y^2 - x) \rightarrow k[y] \otimes_{k[x]} k[z]$ and $\iota_2 : k[x, z] / (z^2 - x) \rightarrow k[y] \otimes_{k[x]} k[z]$,

we get the following commutative diagram:

$$\begin{array}{ccccc}
 k[x, y] & \xrightarrow{i_1} & k[x, y, z] & \xleftarrow{i_2} & k[x, z] \\
 \pi_1 \downarrow & & \vdots & & \downarrow \pi_2 \\
 k[x, y] / (y^2 - x) & & \exists! h & & k[x, z] / (z^2 - x) \\
 & \searrow \iota_1 & \downarrow & \swarrow \iota_2 & \\
 & & k[y] \otimes_{k[x]} k[z] & &
 \end{array}$$

Notice that h is automatically surjective (since the generators of $k[y] \otimes_{k[x]} k[z]$ are $f \otimes 1$ and $1 \otimes g$ in $k[y] \otimes_{k[x]} k[z]$, and such surjectivity onto the generators are given by π_1 and π_2), hence we have $k[y] \otimes_{k[x]} k[z] \cong k[x, y, z] / \ker(h)$.

Which, based on the maps from the two sides, it is natural to see that $y^2 - x, z^2 - x \in \ker(h)$, which we can claim that $(y^2 - x, z^2 - x) = \ker(h)$.

(this part is still in progress).

Final thought: $k[y] \otimes_{k[x]} k[z] \cong k[x, y, z] / (y^2 - x, z^2 - x)$.

6 D

Problem 13

Prove that left adjoint preserves colimit.

Solution: Given a functor $F : \mathcal{C} \rightarrow \mathcal{D}$, and say that F is a left adjoint of functor $G : \mathcal{D} \rightarrow \mathcal{C}$. Let's say a diagram $D : I \rightarrow \mathcal{C}$ (where I is an indexed set) has colimit $K := \lim_{i \rightarrow I} D$ in $\mathcal{C}$. We'll first establish some tools, and gather some statements before proving the main statement:

I. Adjoint Properties on Homsets:

By adjoint property, there exists a natural isomorphism between the two functors $\text{Hom}_{\mathcal{D}}(F(-), -) \cong \text{Hom}_{\mathcal{C}}(-, G(-))$. That is, for any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$, the natural isomorphism provides $\text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$.

II. Cocone functor as Limit in Sets:

Define the functor $\text{Cocone}(-, D) : \mathcal{C} \rightarrow \mathbf{Set}$ as $\text{Cocone}(A, D) = \{\text{Cocones of } D \text{ with tip } A \in \mathcal{C}\}$ (that is, a set collecting all pairs $(A, \{\mu_i : D(i) \rightarrow A\}_{i \in I})$, such that any morphism $f : i \rightarrow j$ in I , satisfies $\mu_i = \mu_j \circ Df : D(i) \rightarrow A$). Then, for any morphism $f : A \rightarrow B$ in $\mathcal{C}$, it defines a morphism $\bar{f} : \text{Cocone}(A, D) \rightarrow \text{Cocone}(B, D)$ satisfying $(A, \{\mu_i : D(i) \rightarrow A\}_{i \in I}) \mapsto (B, \{f \circ \mu_i : D(i) \rightarrow B\}_{i \in I})$.

So, if we consider the functor $\text{Hom}_{\mathcal{C}}(D(-), A) : I^{\text{op}} \rightarrow \mathbf{Set}$, since in $\mathbf{Set}$ all limit exists, in particular we can consider $\lim_{i \in I} \text{Hom}_{\mathcal{C}}(D(i), A) = \{(\mu_i : D(i) \rightarrow A)_{i \in I} \mid \forall f \in \text{Hom}_I(i, j), \mu_i = \mu_j \circ Df\} = L$, where $L \subseteq \prod_{i \in I} \text{Hom}_{\mathcal{C}}(D(i), A)$.

Notice that the set L is precisely all collections of morphisms that's compatible with diagram D , and has target at A . So, $L = \text{Cocone}(A, D)$.

III. $\text{Hom}_{\mathcal{C}}(-, A)$ Sends Colimit to Limit:

Given any diagram $D' : J \rightarrow \mathcal{C}$ such that colimit exists, then let $K' := \lim_{j \in J} D'(j)$ be the colimit (together with morphisms $\{k_j : D'(j) \rightarrow K'\}_{j \in J}$), the relation in **II** provides the following:

$$\text{Hom}_{\mathcal{C}}(K', A) \cong \text{Cocone}(A, D') \cong \lim_{j \in J} \text{Hom}_{\mathcal{C}}(D'(j), A) \quad (6.1)$$

This first set isomorphism is coming from the fact that for each cocone of D' , say $(A, \{\mu_j : D'(j) \rightarrow A\}_{j \in J})$ with tip A , there exists a unique morphism $f_{\mu} : K' \rightarrow A$, such that each $\mu_j = f_{\mu} \circ k_j$. So, there is a natural 1-1 correspondance between cocone with tip A , and the morphisms from K' to A .

Finally, with the above tools, we can prove the statement: Given that K is a colimit of diagram $D : I \rightarrow \mathcal{C}$, then we get the following for all $Y \in \mathcal{D}$, using the Hom functors:

$$\text{Hom}_{\mathcal{D}}(F(K), Y) \cong \text{Hom}_{\mathcal{C}}(K, G(Y)) \cong \lim_{i \in I} \text{Hom}_{\mathcal{C}}(D(i), G(Y)) \cong \lim_{i \in I} \text{Hom}_{\mathcal{D}}(F \circ D(i), Y) \quad (6.2)$$

Where the first isomorphism is using **I**, the second isomorphism uses **III**, and the third isomorphism uses **I** again.

The last step is that $\lim_{i \in I} \text{Hom}_{\mathcal{D}}(F \circ D(i), Y) \cong \text{Cocone}(Y, F \circ D)$ based on **II**. Hence, for all $Y \in \mathcal{D}$, one has $\text{Hom}_{\mathcal{D}}(F(K), Y) \cong \text{Cocone}(Y, F \circ D)$, which is precisely the characterization of a colimit of $F \circ D : I \rightarrow \mathcal{D}$ (Since $F(K)$ is a colimit of $F \circ D \iff$ every cocone of $F \circ D$ of tip Y corresponds to a unique morphism $F(K) \rightarrow Y \iff \text{Hom}_{\mathcal{D}}(F(K), Y) \cong \text{Cocone}(Y, F \circ D)$).

So, $F(K)$ is a colimit of $F \circ D$, this shows that F (as a left adjoint of G) preserves colimit.

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Problem 14

Give an example that tensor product is not left exact in general.

Solution: Consider the $\mathbb{Z}$ -module homomorphism $m : \mathbb{Z} \hookrightarrow \mathbb{Z}$ by $m(a) = 2a$ for all $a \in \mathbb{Z}$ (where m is injective). Then, it since $\text{im}(m) = 2\mathbb{Z}$, this indicates that a cokernel of m can be given by $\pi : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ via canonical projection. Which, it forms the short exact sequence $0 \rightarrow \mathbb{Z} \hookrightarrow \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ (the second map is m , the third map is π).

However, if consider the tensor product of the above ones with $\mathbb{Z}$ -module $\mathbb{Z}/2\mathbb{Z}$, then the form the map $1 \otimes m : \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}$ by $(1 \otimes m)(a \otimes b) = a \otimes (2b)$ for all $a \in \mathbb{Z}/2\mathbb{Z}$ and $b \in \mathbb{Z}$. Yet, notice that this map is a zero map, since $a \otimes (2b) = (2a) \otimes b = 0 \otimes b = 0$ (because $2a = 0 \in \mathbb{Z}/2\mathbb{Z}$, where a is arbitrary). This shows that an injective module homomorphism after tensoring with some modules (and the corresponding identity map), it doesn't necessarily preserve injectivity (in this case, it turns m an injective morphism to $1 \otimes m$, a zero map on nonzero module, since $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ when tensor as $\mathbb{Z}$ -modules).

Since this tensor product with $\mathbb{Z}/2\mathbb{Z}$ doesn't necessarily preserve injectivity, it doesn't preserve the left side of short exact sequence (since $0 \rightarrow \mathbb{Z} \hookrightarrow \mathbb{Z}$ by m , gets send to $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}$ a zero map, which is no longer injective). So, tensor with $\mathbb{Z}/2\mathbb{Z}$ is not left exact, which serves as a counterexample here.